

Modified Dual-Stage Controller for Dealing With Secondary-Stage Actuator Saturation

Guoxiao Guo, *Member, IEEE*, Daowei Wu, and Tow Chong Chong, *Member, IEEE*

Abstract—This paper proposes a modified decoupled master-slave (DMS) dual-stage control scheme for hard disk drives that offers enhanced stability when the secondary-stage actuator is saturated. A nonlinear secondary-stage actuator model is used instead of a linear model in the controller. This modification allows microactuators with relatively smaller displacement range to be used in the servo system without causing stability problems in the DMS controller. Simulation and experimental results confirm the effectiveness and robustness of the scheme.

Index Terms—Control saturation, dual-stage servo, hard disk drive servo, saturation nonlinearity.

I. INTRODUCTION

DUAL-STAGE actuation has long been recognized as a powerful candidate to support higher track densities in hard disk drives (HDDs) because of its ability to provide higher bandwidth actuation and, thus, to reject a wider range of vibration. Future storage densities of over 1 Tb/in² will require track densities of over 540 000 tracks per inch (TPI), which cannot be supported by single-stage actuation alone [1], [2].

A few types of secondary-stage actuators or simply microactuators, namely, actuated suspension [3], actuated slider [4]–[6], and actuated head [7], have been studied over the years. The actuated suspension, which is simpler in structure, relatively easier to implement, and sometimes called a milliactuator, has been shown to support a servo bandwidth of about 3 kHz [8]. The actuated sliders and actuated heads place the read/write (R/W) element and the actuator closer together, have a smaller mass to actuate, and thus enable higher servo bandwidths compared with actuated suspensions. However, the range of displacement of these microactuators is typically smaller because the size of the displacement amplification mechanism is limited by the slider size [6]. Nevertheless, the voice coil motor (VCM) and microactuator movement might conflict with each other at certain frequencies in dual-stage loops [4]–[11]. Thus, the microactuators are typically required to cover more than one track's displacement range to avoid actuator saturation, although the track following servo only needs to maintain the read/write head within a fraction of the track width. On the other hand, even if the microactuator has a larger than one track's displacement range which can help the improvement of settling process ([10]), due to the large positioning error at the point the dual-stage loop is closed, the microactuator still tends to saturate. It is well known that actuator saturation causes stability problems in

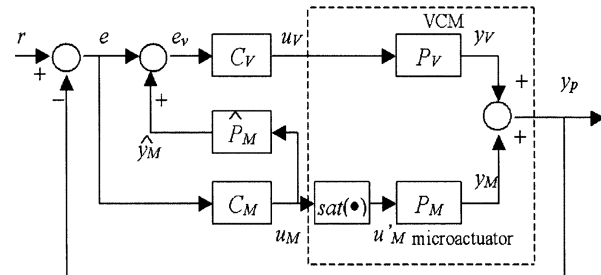


Fig. 1. Dual-stage control.

control systems. Thus, ensuring the stability of the dual-stage loop with a simple yet effective control method is necessary for the dual-stage control system that might have a saturating secondary-stage actuator [12], [13].

In this paper, we study the stability enhancement of the decoupled master-slave (DMS) dual-stage control configuration when the microactuator allows a limited driving voltage. Instead of following the traditional approaches dealing with actuator saturation (see [13]–[17] and the references therein), we exploit the special DMS structure to provide a simple solution [12]. We show that by simply using the nonlinear plant model in the observer design recommended in control textbook (see, e.g., [18]), the stability of the closed-loop system can be greatly improved with a saturating secondary-stage actuator. Both experimental and simulation results show that the design is robust against inaccurate microactuator parameters.

II. NONLINEAR CONTROL SCHEME FOR DEALING WITH SECONDARY-STAGE ACTUATOR SATURATION

The dual-stage servo uses a secondary-stage actuator piggy-back on the VCM. As such, its control system block diagram can be represented as Fig. 1, in which the double-input single output (DISO) plant is shown in the dashed line block. The control scheme is the DMS configuration whereby the model $\hat{P}_M$ estimates the microactuator output, and the controllers C_V and C_M generate two control outputs for the two actuators P_V (VCM) and P_M (microactuator), respectively. Controllers for such a DISO system would otherwise be easy to design had we not required the two nonideal actuators closely cooperate with each other for optimal performance.

Because the two actuators are supposed to work closely together, how the microactuator loop and VCM loop cross each other becomes one of the major concerns in the design. Typically, the VCM is required to follow signals of up to a few hundred hertz, whereas the microactuator with limited

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displacement range works to track higher frequency signals. The two actuators hand off at about a few hundred hertz [9].

In the usual DMS dual-stage control scheme, if $\hat{P}_M = P_M$, it is well known that the closed-loop transfer function T (from r to y_p) and the sensitivity transfer function $S = 1 - T$ are

$$T = \frac{C_V P_V + C_M P_M + C_V P_V C_M P_M}{(1 + C_V P_V)(1 + C_M P_M)} \quad (1)$$

$$S = \frac{1}{(1 + C_V P_V)(1 + C_M P_M)} \quad (2)$$

respectively, if the actuators are not saturated. The sensitivity transfer functions for the fictitious VCM-only loop (assuming $C_M = 0$) S_V and microactuator-only loop (assuming $C_V = 0$) S_M are

$$S_V = \frac{1}{1 + C_V P_V} \quad (3)$$

$$S_M = \frac{1}{1 + C_M P_M} \quad (4)$$

respectively. Noting that

$$S = S_V S_M \quad (5)$$

the servo designer thus can select C_V and C_M independently to shape the combined sensitivity transfer function S .

If we define

$$C_{VP} = C_V(1 + C_M \hat{P}_M) \quad (6)$$

where C_{VP} is the VCM controller for the parallel implementation scheme, the DMS structure can be implemented in a parallel way. Furthermore, the VCM loop and microactuator hand off at the frequency whereby $C_{VP} P_V$ and $C_M P_M$ have the same amplitude. The controller parameters can be selected using classical loop shaping or optimal control for the given control structures (see for example [4], [9]–[11] and the references therein) to achieve optimality, robustness, and implementability requirements.

Although the control design methods have widely been discussed in HDD servo literatures, these designs typically assumed a nonsaturating actuator. However, to protect the microactuator from being damaged, typically the control signal applied to the microactuator is limited

$$u'_M = \text{sat}(u_M) = \begin{cases} \bar{u}_M, & \text{if } u_M \geq \bar{u}_M \\ u_M, & \text{if } \underline{u}_M < u_M < \bar{u}_M \\ \underline{u}_M, & \text{if } u_M \leq \underline{u}_M \end{cases} \quad (7)$$

where $\underline{u}_M$ and $\bar{u}_M$ represent the minimum and maximum allowable control voltage to the microactuator.

Based on the DMS control structure, we next discuss a modified DMS scheme. Fig. 2 shows the proposed modified DMS scheme, which has a block $\text{sat}(\cdot)$ in the controller compared with the usual DMS scheme shown in Fig. 1. To see how incorporating the microactuator model with saturation in the DMS controller helps to improve the stability of the control loop against the microactuator saturation, assume that $r = 0$, and the transfer function of a saturation section is represented

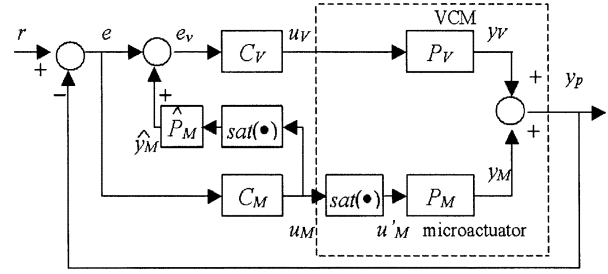


Fig. 2. Dual-stage control considering microactuator saturation.

as $\text{sat}(\cdot)$. By cutting the VCM loop open at the point e_V in Fig. 2, we have the VCM path open-loop transfer function

$$O_V = C_V P_V \left(1 + C_M \hat{P}_M \text{sat}(\cdot) \right) \frac{1}{1 + C_M P_M \text{sat}(\cdot)}. \quad (8)$$

Obviously, if $\hat{P}_M = P_M$, and assuming that the microactuator P_M and its model $\hat{P}_M$ have the same initial values, we have $\hat{y}_M(k) = y_M(k)$, resulting effectively in

$$\left(1 + C_M \hat{P}_M \text{sat}(\cdot) \right) \frac{1}{1 + C_M P_M \text{sat}(\cdot)} = 1. \quad (9)$$

Consequently, the secondary-stage actuator model does not appear in the VCM path; thus, its saturation will not affect the VCM control loop's stability.

In contrast, in the usual linear control configuration shown in Fig. 1, we have $|\hat{y}_M(k)| \gg |y_M(k)|$ if the microactuator is saturated for a sufficiently long time. In this case, the effective open-loop transfer function

$$O_V = C_V P_V (1 + C_M \hat{P}_M) \frac{1}{1 + C_M P_M \text{sat}(\cdot)} \quad (10)$$

will have a higher effective gain than (8), leading to potential instability.

It could be noted that for MEMS-based microactuators with relative position error signal ([11]), y_M is available and, thus, can be used in the control scheme directly instead of $\hat{y}_M$. In case y_M is not available, its estimation $\hat{y}_M$ can be used as in Fig. 2. In general, an open-loop estimator does not guarantee asymptotic stability of the error dynamics [18]. However, we can verify (see the Appendix) that if P_M is stable, and $\hat{P}_M = P_M$, we have

$$\lim_{k \rightarrow \infty} \hat{y}_M(k) - y_M(k) = 0. \quad (11)$$

We will see through an example in the next section that even for reduced-order PZT actuated suspension with parameter uncertainty, $\hat{y}_M$ is still a close estimation of y_M , and hence the modified algorithm is still effective in dealing with microactuator saturation.

III. EXPERIMENTAL RESULTS

Although the method shown in the previous section applies to various types of dual-stage actuators, we will use an actuated suspension to show the effectiveness of the proposed nonlinear control scheme against the original linear control scheme in this section.

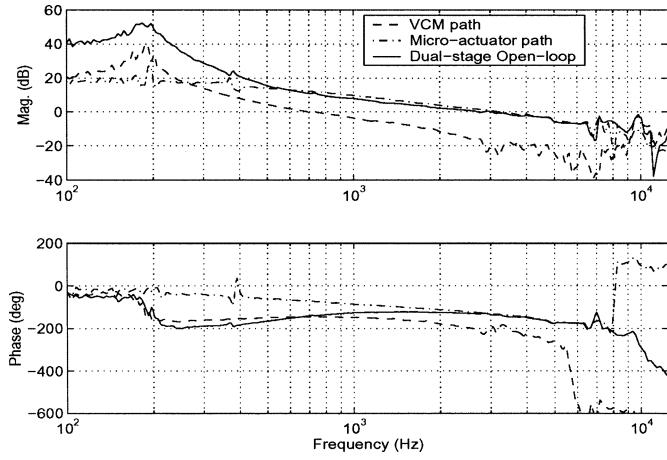


Fig. 3. Bode plots.

The nominal control loops, i.e., VCM path, microactuator path, and overall dual-stage open loop are shown in Fig. 3 with

$$P_V(s) = \frac{2.04 \times 10^{21}}{s + 3.14 \times 10^4} \frac{1}{s^2 + 2073s + 4.3 \times 10^8} \times \frac{1}{s^2 + 1508s + 3.55 \times 10^8} \times \frac{1}{s^2 + 1244s + 1.7 \times 10^9} \mu\text{m/V} \quad (12)$$

$$P_M(s) = \frac{20 \times 5.45 \times 10^7}{s^2 + 2450s + 1.7 \times 10^9} \times \frac{s^2 + 4524s + 2.08 \times 10^9}{s^2 + 6032s + 3.64 \times 10^9} \mu\text{m/V}. \quad (13)$$

$$C_V(z) = \frac{1.6(z - 0.9875)^2}{(z - 0.2846)(z - 0.99)} \frac{z^2 - 1.416z + 0.9415}{z^2 - 1.297z + 0.92} \times \frac{z^2 + 0.1525z + 0.95}{z^2 + 0.4155z + 0.0144} \text{ V/V} \quad (14)$$

$$C_M(z) = \frac{1.227(z - 0.081)}{z - 0.9158} \frac{z^2 + 1.32z + 0.7856}{z^2 + 0.6232z - 0.277} \times \frac{z^2 + 0.148z + 0.9}{z^2 + 0.439z + 0.84} \text{ V/V}. \quad (15)$$

We note here that the plant models were measured with a laser Doppler vibrometer (LDV) at $2 \mu\text{m/V}$; the microactuator driver has a gain of 20; the controllers were implemented in a dSPACE 1103 system with a sampling frequency of 25 kHz. The VCM controller $C_V(z)$ includes two notch filters to cancel the VCM resonances around 3.0 and 6.5 kHz. For the compensated VCM model, a lead-lag compensator is used for the desired gain and phase margins. Because the microactuator has less phase loss than the VCM, its controller $C_M(z)$ is a lag compensator in series with two notch filters for microactuator resonances at 6.5 and 9.6 kHz to achieve the target open-loop bandwidth.

A. Step Responses

To show that the proposed method is effective in dealing with secondary-stage actuator with input saturation, we restricted the input of the microactuator to no more than $\pm 0.15 \text{ V}$ which caps the microactuator output to $\pm 0.12 \mu\text{m}$, or $\pm 4.8 \mu\text{in}$ at steady state.

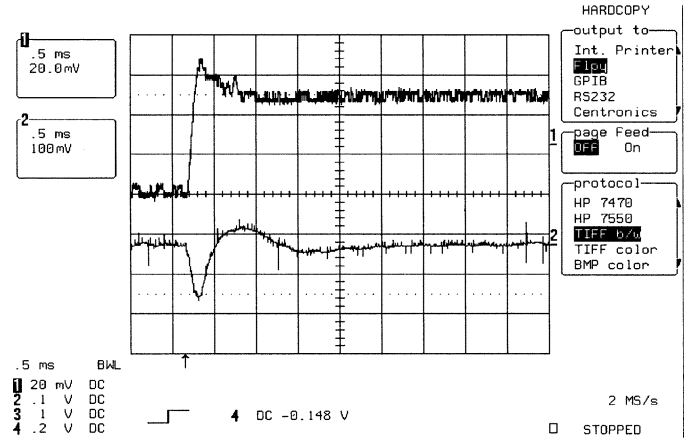
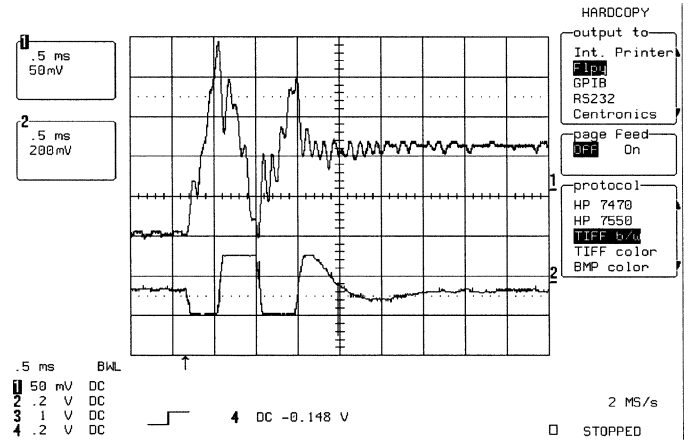
Fig. 4. $4\text{-}\mu\text{in}$ step response. Upper curve: combined output y_p . Lower curve: microactuator control voltage u'_M .Fig. 5. Linear control $10\text{-}\mu\text{in}$ step response. Upper curve: combined output y_p . Lower curve: microactuator driving voltage u'_M .

Fig. 4 shows a $4\text{-}\mu\text{in}$ step response of the dual-stage controller. The microactuator is not saturated, so the linear controller and the nonlinear controller behave the same.

Fig. 5 shows the linear controller response with a $10\text{-}\mu\text{in}$ step input. The position output is oscillatory, and the settling time is as long as 1.5 ms. Further increasing the reference signal amplitude causes the dual-stage loop to become unstable. When the proposed controller is used, however, the dual-stage loop is still stable with a step of $35 \mu\text{in}$ (Fig. 6) or higher.

B. Effect of Saturation Level

Fig. 7 shows the comparison of 3σ for y_p with the secondary actuator saturation range set at $\pm 0.05, \pm 0.1, \pm 0.15, \pm 0.2, \pm 0.25, \pm 0.3, \pm 0.4, \pm 0.5$, and $\pm 0.7 \text{ V}$ in our experiment. As can be seen from the figure, the 3σ for the nonlinear controller degraded from 0.5 to $1.1 \mu\text{in}$ when the microactuator control voltage limit reduced from ± 0.7 to $\pm 0.05 \text{ V}$. The linear controller, while achieving the almost same 3σ of $0.5 \mu\text{in}$ at a control voltage range of $\pm 0.7 \text{ V}$, could not maintain the closed-loop stable when the control voltage was reduced to slightly below $\pm 0.05 \text{ V}$.

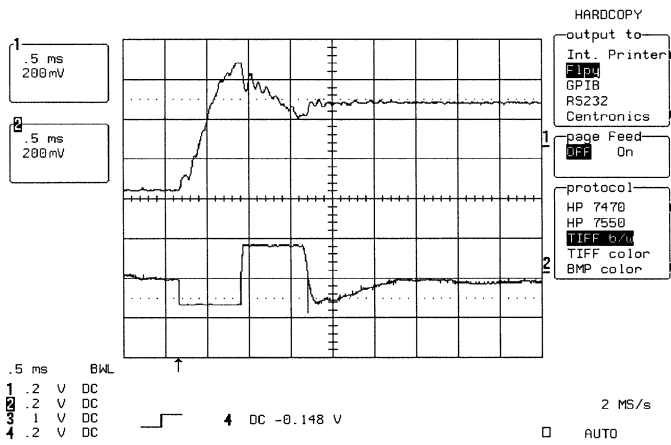


Fig. 6. Nonlinear control 35- μ in step response. Upper curve: combined output y_p . Lower curve: microactuator driving voltage u'_M .

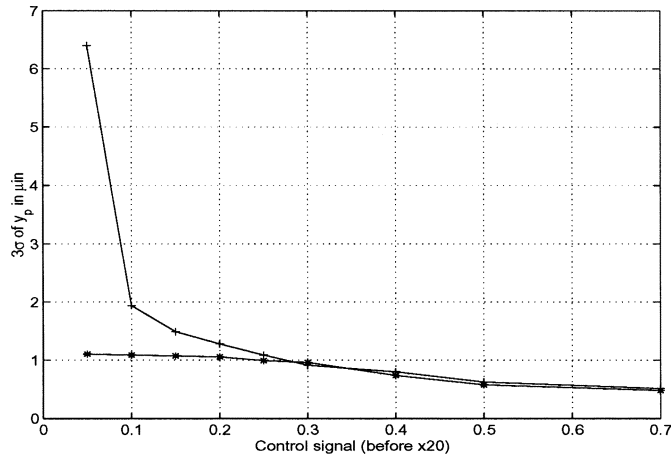


Fig. 7. Combined output 3σ versus saturation range experimental results. (+) Linear controller. (*) Nonlinear controller.

C. Robustness Evaluation

Noting that $\hat{y}_M \rightarrow y_M$ requires an accurate microactuator model, to evaluate the robustness of the proposed algorithm, Fig. 8 shows the step responses of the system with different step inputs with a constant $\hat{P}_M$ equal to the dc gain of P_M . The microactuator control voltage is limited to ± 0.15 V. Fig. 9 shows the corresponding $\hat{y}_M$ and y_M . As can be seen from the figures, with a pure gain estimating the microactuator output, the proposed scheme still retains a reasonably good stability in the face of microactuator saturation.

To further investigate the robustness when using reduced-order microactuator model, Table I summarizes the maximum allowable step changes (MASC) with a stable closed loop for three cases obtained through simulation. Case 1 uses (14) and (15) to control the plant given in (12) and (13) using the block diagram shown in Fig. 1. Case 2 uses the full-order $\hat{P}_M$ for the proposed modified DMS scheme. Case 3 uses the modified DMS scheme, but $\hat{P}_M$ is only the dc gain of P_M . Fig. 10 plots the MASC versus percent dc gain error for the linear control scheme. It shows that the MASC decreases from 28 to 7.6 μin as the dc gain error varies from -50% to $+50\%$. For the modified nonlinear scheme with full-order and reduced-order models

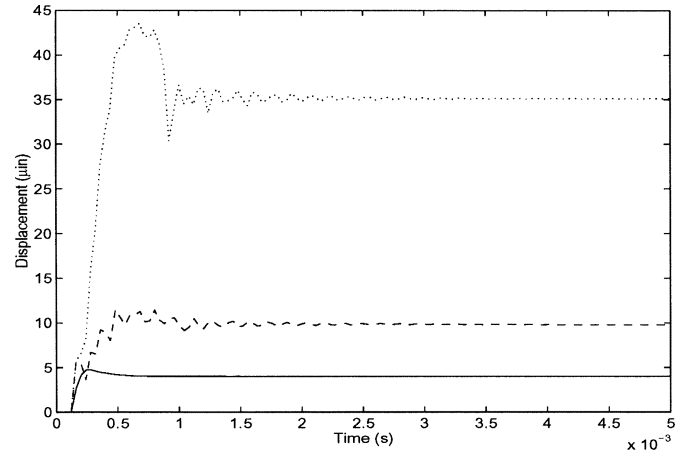


Fig. 8. Combined output responses with reduced-order microactuator models. $r = 4$ μin (solid), 10 μin (dashed), and 35 μin (dotted).

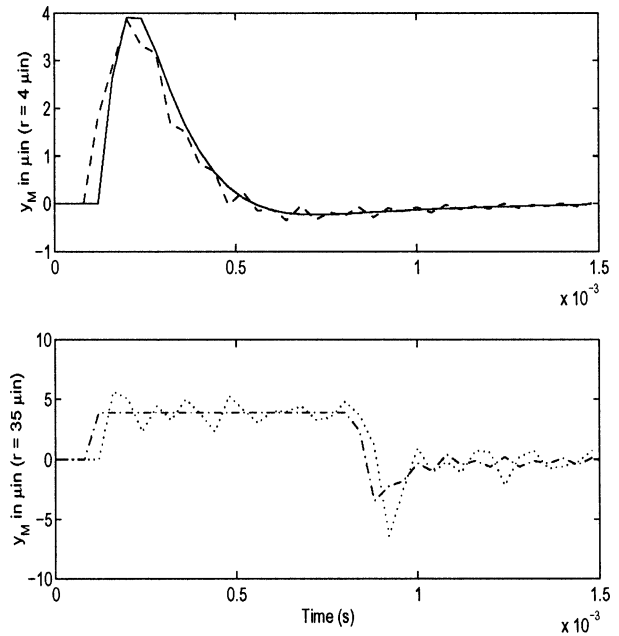


Fig. 9. Microactuator and estimator responses with reduced-order microactuator models for $r = 4$ μin (y_M : solid line, $\hat{y}_M$: dashed line) and 35 μin (y_M : dotted line, $\hat{y}_M$: dash-dotted line).

TABLE I
MASC FOR DIFFERENT CONTROL SCHEMES

Control scheme	MASC in μin
Case 1	10
Case 2	> 1000
Case 3	> 1000

for $\hat{P}_M$, the MASC can still be greater than 1000 μin when the dc gain of $\hat{P}_M$ fluctuates in between $\pm 50\%$ of its nominal value.

It should be noted that in Fig. 10, a lower value for $\hat{P}_M$, although increasing the MASC, will decrease the gain of the VCM path, and affect the hand-off of the dual-stage loops even if the microactuator is not saturated. The proposed nonlinear modification, on the other hand, has a much larger MASC with a saturating secondary-stage actuator and retains all the properties of the original linear design when the microactuator is not saturated.

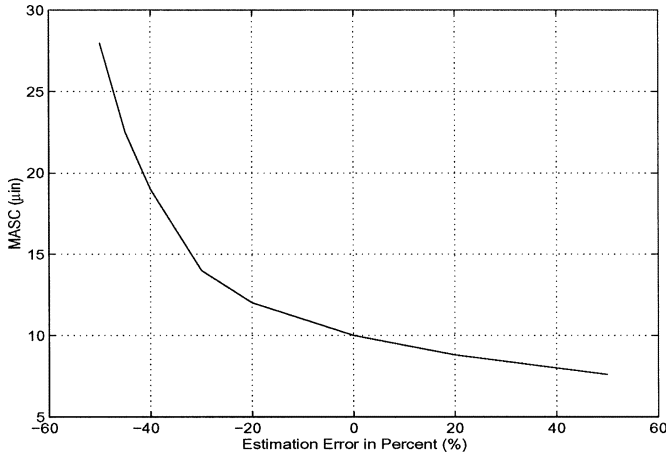


Fig. 10. MASC for linear scheme with reduced-order microactuator model.

IV. CONCLUSION

This paper presents a simple modification to the decoupled master-slave dual-stage control configuration, which greatly improved the servo loop performance and stability when the secondary-stage actuator saturates. A nonlinear secondary-stage actuator model is used to predict its output. Our testing results show that the proposed nonlinear controller retains all the properties of the original linear design when the secondary-stage actuator is not saturated; when the secondary-stage actuator is saturated, the proposed design can enhance the stability of the dual actuator servo loop. The design is robust against the microactuator model uncertainties. Additionally, the design method can be applied to other dual-actuator systems with a saturating secondary-stage actuator, such as power systems with flexible ac transmission devices [19].

APPENDIX

To verify (11), assume that the microactuator P_M is represented by state space equations

$$\begin{cases} x_M(k+1) = F_M x_M(k) + \Gamma_M u'_M(k), \\ y_M = H_M x(k) \end{cases} \quad (16)$$

and the estimator $\hat{P}_M$ by

$$\begin{cases} \hat{x}_M(k+1) = F_M \hat{x}_M(k) + \Gamma_M u'_M(k), \\ \hat{y}_M = H_M \hat{x}(k). \end{cases} \quad (17)$$

The error dynamics of the estimator is given by

$$\begin{aligned} \bar{x}(k+1) &= x(k+1) - \hat{x}(k+1), \\ &= F_M \bar{x}(k). \end{aligned} \quad (18)$$

Since the microactuator model is assumed stable, so is the error dynamics (18). Thus, we have

$$\lim_{k \rightarrow \infty} y_M(k) - \hat{y}_M(k) = \lim_{k \rightarrow \infty} H_M \bar{x}(k) = 0. \quad (19)$$

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